

No Universal Bound of κ_d by β

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Status and source

This packet records a candidate full negative answer, likely valid, to Question 1 of Gonzalez and Martinon, *A quantitative approach to disjointly non-singular operators*, arXiv:2108.01052 [1]. The question appears in Section 7.1 on PDF page 11.

7.1. Open Questions. We finish the paper stating some open questions.

Question 1. *Is $\kappa_d \leq D \cdot \beta$ for some constant $D > 0$?*

The notation in the published PDF is transcribed as follows:

Question 1. Is $\kappa_d \leq D \cdot \beta$ for some constant $D > 0$?

The answer is no. The counterexample is an elementary strictly singular off-diagonal perturbation of the identity on the order-continuous atomic Banach lattice $\ell_1 \oplus_\infty \ell_2$.

Definitions and two source facts

For a Banach lattice E , a Banach space Y , and $T \in \mathcal{L}(E, Y)$, write $d(E)$ for the set of disjoint sequences of nonzero vectors and $j(A) = \inf_{\|x\|=1} \|Ax\|$. The quantities in the question are

$$\tau_d(T) = \sup_{(x_n) \in d(E)} j(T|_{[x_n]}), \quad \kappa_d(T) = \inf_{(x_n) \in d(E)} \tau_d(T|_{[x_n]}),$$

and

$$\beta(T) = \inf \left\{ \liminf_{n \rightarrow \infty} \|Tx_n\| : (x_n) \subset S_E \text{ is disjoint} \right\}.$$

We also use the operational quantity

$$\Delta_d(R) = \sup_{(x_n) \in d(E)} \inf_{(y_n) \in d([x_n])} \|R|_{[y_n]}\|.$$

Two results proved in the source paper are:

1. R is disjointly strictly singular if and only if $\Delta_d(R) = 0$;
2. for $A, B \in \mathcal{L}(E, Y)$,

$$\kappa_d(A + B) \leq \kappa_d(A) + \Delta_d(B).$$

These are Propositions 4.4 and 5.5(2), respectively, in the published paper. The second inequality follows directly from $j(A + B) \leq j(A) + \|B\|$ and the two nested disjoint-sequence operations. In particular, adding a disjointly strictly singular operator does not change κ_d : apply the inequality in both directions.

Construction

Let

$$E = \ell_1 \oplus_\infty \ell_2, \quad \|(x, y)\|_E = \max\{\|x\|_1, \|y\|_2\}.$$

With the coordinatewise order, E is an order-continuous atomic Banach lattice. Let $J : \ell_1 \rightarrow \ell_2$ be the formal inclusion $J(x_n) = (x_n)$.

Lemma 1. *The operator $J : \ell_1 \rightarrow \ell_2$ is strictly singular.*

Proof. Suppose that J were bounded below on an infinite-dimensional closed subspace M of ℓ_1 . Then $J(M)$ would be a closed subspace of ℓ_2 , and $J|_M$ would be an isomorphism from M onto $J(M)$. Consequently M would be reflexive. On the other hand, every subspace of ℓ_1 has the Schur property. A reflexive Schur space is finite-dimensional: its unit ball is weakly compact, hence weakly sequentially compact, while the Schur property converts weakly convergent sequences into norm-convergent sequences. This contradiction proves the claim. $\square$

For $c > 1$, define

$$S_c(x, y) = (0, cJx), \quad T_c = I_E - S_c,$$

so explicitly

$$T_c(x, y) = (x, y - cJx).$$

Lemma 2. *S_c is strictly singular, $S_c^2 = 0$, and T_c is invertible with $T_c^{-1} = I_E + S_c$.*

Proof. The nilpotence and inverse formula are immediate. To prove strict singularity, suppose that S_c is bounded below by $\delta > 0$ on an infinite-dimensional closed subspace $M \subset E$. For $z = (x, y) \in M$,

$$\delta\|z\|_E \leq \|S_c z\|_E = c\|Jx\|_2 \leq c\|x\|_1.$$

Thus the first-coordinate projection is bounded below on M , so its image $M_1 \subset \ell_1$ is a closed infinite-dimensional subspace. Moreover, for x arising as the first coordinate of $z \in M$,

$$c\|Jx\|_2 = \|S_c z\|_E \geq \delta\|z\|_E \geq \delta\|x\|_1.$$

Hence J is bounded below on M_1 , contradicting Lemma 1. $\square$

The full negative result

Theorem 1. *For every $c > 1$, the invertible operator $T_c \in \mathcal{L}(E)$ satisfies*

$$\kappa_d(T_c) = 1, \quad \frac{1}{1+c} \leq \beta(T_c) \leq \frac{1}{c}.$$

Consequently

$$\frac{\kappa_d(T_c)}{\beta(T_c)} \geq c,$$

and there is no universal constant $D > 0$ for which $\kappa_d(T) \leq D\beta(T)$ holds for all Banach lattices and operators.

Proof. Since S_c is strictly singular, it is disjointly strictly singular and therefore $\Delta_d(S_c) = 0$. The perturbation inequality gives

$$\kappa_d(T_c) = \kappa_d(I_E - S_c) \leq \kappa_d(I_E) + \Delta_d(-S_c) = 1.$$

Applying the same inequality to $I_E = T_c + S_c$ gives

$$1 = \kappa_d(I_E) \leq \kappa_d(T_c) + \Delta_d(S_c) = \kappa_d(T_c).$$

Thus $\kappa_d(T_c) = 1$.

Let (e_n) denote the usual unit vectors in both ℓ_1 and ℓ_2 , and put

$$u_n = (c^{-1}e_n, e_n) \in E.$$

The vectors u_n are normalized and pairwise disjoint. Direct computation gives

$$T_c u_n = (c^{-1}e_n, 0), \quad \|T_c u_n\|_E = c^{-1}.$$

Therefore $\beta(T_c) \leq c^{-1}$.

Finally, $T_c^{-1} = I_E + S_c$ and $\|T_c^{-1}\| \leq 1 + c$, so

$$\beta(T_c) \geq j(T_c) \geq \frac{1}{1+c} > 0.$$

The ratio estimate follows. Given a proposed universal D , choose $c > D$; then

$$D\beta(T_c) \leq D/c < 1 = \kappa_d(T_c),$$

which disproves the proposed inequality. □

Remark 1. Both κ_d and β are homogeneous. Thus, if a norm-one family is desired, replace T_c by $T_c/\|T_c\|$; the ratio $\kappa_d(T_c)/\beta(T_c)$ is unchanged. The counterexamples are not merely DN-S: every T_c is an isomorphism, and the domain is order continuous.

Why the construction works

The graph vectors $(c^{-1}e_n, e_n)$ make the off-diagonal term cancel their entire ℓ_2 component, so β sees images of size c^{-1} . In contrast, κ_d is stable under disjointly strictly singular perturbations. Thus the same off-diagonal term is invisible to κ_d , even though it produces large pointwise cancellation on a specific normalized disjoint sequence. This is exactly the minimax gap that a universal blocking argument would have had to eliminate.

Verification and novelty notes

An adversarial step check is included as `verification_report.md`. Its verdict is *likely valid*. The quantifiers, disjointness calculation, strict singularity, invertibility, and both operational-quantity inequalities were checked separately.

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2108.01052 and the exact κ_d - β comparison. A bounded web search located the source, later adjacent DN-S work, and the classical operational-quantity paper, but no explicit later answer to Question 1. This is not a literature-complete novelty claim. The triangular strictly singular perturbation mechanism is classical in spirit; the claim here is the explicit application to the source question.

Human review recommendation

Send to a Banach-space/operator-ideal expert as a candidate full counterexample. The highest-value checks are the published convention for Δ_d and Proposition 5.5(2), followed by the fact that strict singularity implies disjoint strict singularity. No unproved geometric claim remains in the construction.

References

- [1] M. Gonzalez and A. Martinon, *A quantitative approach to disjointly non-singular operators*, Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Mat. RACSAM **115** (2021), Article 185; arXiv:2108.01052; DOI: 10.1007/s13398-021-01124-1.